

Annexure III: Teaching Strategy

CLO No.	Teaching-Learning Activities for Active Learning	Assessment (Formative and Summative) Activities
CLO 1	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Practical sessions	1. Continuous Assessment 2. Practical exam 3. Assignment
CLO 2	1. Discussion 2. Reflection 3. Assignment on topics 4. Practical Sessions 5. Think Pair Share	1. Continuous Assessment 2. Practical exam 3. Assignment
CLO 3	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Practical Sessions	1. Continuous Assessment 2. Practical exam 3. Assignment
CLO 4	1. Problem Solving 2. Assignment on topics 3. Practical Sessions	1. Continuous Assessment 2. Practical exam 3. Assignment

Annexure IV: Course Outline

Institute	Institute of Technology
Department	Computer Science and Engineering
Programme	B. Tech. (CSE), Int. B. Tech. CSE-MBA
Batch	2024
Term/Semester	5 th
Course Title	Computer Networks
Course Code	3CS201CC24
Credit Hours	L-3+P-2
Faculty Members	Dr Pooja Chaturvedi(Course Coordinator) Dr Vijay Ukani Dr Zunnun Narmawala Dr Umesh Bodkhe Dr Jigna Hathaliya Dr Rajan Datt
E-mail ID (CC)	pooja.chaturvedi@nirmauni.ac.in
Contact No. (CC)	09918596994
Office Hours (CC)	Monday to Friday: 7:45 am to 3:00 pm Sat: 7:30 am to 1:30 pm Check Timetable for availability (https://nuweb.nirmauni.ac.in/tt/cse)
Moodle Details	

a. Course Overview

The entire world has become very small thanks to the evolution of telecommunication and networking. We use a large number of communication applications. E-mail, Whatsapp and Facebook are a few of those. As a student of Computer Engineering, one must know the internal details of how applications/devices communicate. The objective of this course is the same. This course discusses issues and protocols involved in packet transfer over a network or Internet.

This course will prepare students for careers in network administration, network engineering, cybersecurity, and other IT-related fields. By the end of the course, students will

have a solid foundation in networking principles and the practical skills needed to design, implement, and manage modern computer networks.

b. Course Learning Outcomes (CLOs)

At the end of the course, the students shall be able to:

1. summarise the functionalities of different layers of computer network architectures (BL2)
2. analyse protocols related to various network architecture layers (BL4)
3. simulate various protocols for different types of networks (BL5)
4. design computer network configurations (BL6)

c. Text Book (reflect other study/reference materials in session plan)

1. Andrew S. Tanenbaum, Computer Networks, PHI Publication
2. Behrouz Forouzan, Data Communication Networking, TMH Publication
3. Behrouz Forouzan, TCP/IP Protocol suite, TMH Publication
4. William Stallings, Data and Computer Communication, Pearson
5. Jim Kurose, Computer Networking: A top down approach, Pearson

d. Assessment Components & Schedule

SEE	CE	LPW
1.0	1.0	1.0

Component (with No. of sub-components)	Weightage (each part of the Component)	Schedule	Marks	CLO
Class Test	20% of CE	To be announced	20	1,2,3, and 4
Sessional Test	60% of CE	08/09/2026 to 11/09/2026	50	1,2,3, and 4
Term Paper/Assignments	20% of CE	30/10/2026	20	1,2,3, and 4
SEE	100% of SEE	19/11/2026 onwards	100	1,2,3 and 4

LPW Exam	40% of LPW	23/10/2026 to 30/10/2026	40	1,2,3 and 4
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e. Session Plan

Sess. No.	Topic/Sub Topic	Session Details	
1	Introduction to the course, reading materials and evaluation methods	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		Course learning outcome (CLO)	1
		Session Learning Outcome (SLO)	Course basics
		URL	-
2	Network Hardware: Personal Area Networks, Local Area Networks, Metropolitan Area Networks, Wide Area Networks, Internetworks Network Software: Protocol Hierarchies, Design Issues of Layers, Connection Oriented v/s Connectionless services, Service Primitives	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1
		SLO	Overview of Network Hardware and Software
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
3	OSI Reference Model: Description of all 7 layers TCP/IP Reference Model, Comparison of OSI and TCP/IP	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1

		SLO	Knowledge of Network Architectures
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
4	Example Networks: Internet, Cellular Networks, Wireless LANs, RFID and Sensor Networks Network Standardization, Revision of First Unit of Syllabus	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1
		SLO	Understanding of Types of Networks
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
5	Data Link Layer Design Issues: Services Provided to Network Layer, Framing, Error Control, Flow Control	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1
		SLO	Framing, Error Control and Flow Control and Data Link Layer
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
6	Elementary Data Link Protocols	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading(Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Basics of Data Link Layer Protocols
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
7	Simplex Stop and Wait for Noiseless and Noisy Channels	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading(Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2

		SLO	Simplex Stop and Wait Protocol Design
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
8	Sliding Window Protocols: One bit	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading(Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1
		SLO	One-bit sliding window protocol design
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
9	Go back N, Selective Repeat protocols	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Go back N and Selective Repeat Protocols' design
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
10	Example Data Link Protocols: Packet over SONET, ADSL	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Basics to SONET and ADSL protocols
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
11		Text Book	Computer Networks by Andrew S Tanenbaum

	Protocol Verification Models	Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	How to verify protocol design
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
12	Static and Dynamic Channel Allocation, ALOHA and CSMA	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Basics of static and dynamic channel allocation
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
13	Collision-Free Protocols, Limited Contention Protocols, Wireless LAN Protocols	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Understanding of collision-free protocols
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
14	Ethernet: Classical Ethernet Physical Layer, Ethernet MAC sublayer protocol	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Knowledge of Ethernet physical and MAC sublayer
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
15	Ethernet Performance, Switched Ethernet, Fast Ethernet, Gigabit Ethernet	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading(Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2

		SLO	Analysis of Ethernet performance and understanding of types of Ethernet
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
16	Wireless LANs: 802.11 standard architecture, protocol stack	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Basics of Wireless LANs
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
17	MAC protocol, Frame structure	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	1,2
		SLO	Frame Structures of MAC protocols
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
18	Data Link Layer Switching: Use of bridges, learning bridges, spanning tree bridges	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3
		SLO	Data Link Layer Switching basics and knowledge of types of bridges
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
19	Hubs, Switches, Routers, gateways, Virtual LANs	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3
		SLO	Difference between devices at different layers and working of Virtual LANs

		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
20	Network Layer Design Issues: Store and Forward Concept, Services, Comparison of Virtual Circuit and Datagram subnet	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3
		SLO	Understanding of store and forward concept, Virtual Circuit and Datagram Subnet
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
21	Shortest Path Routing, flooding	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Details of Shortest path routing, Working of Flooding approach
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
22	Distance Vector Routing: Principle, count to infinity problem and its solutions	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Understanding of Distance Vector Routing and its challenges
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
23	Link State Routing: Working Principle	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Understanding of Link State Routing and its working principle
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446

24	Broadcast, Multicast and Anycast routing	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Working of different types of routing
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
25	Congestion Control Algorithms: Traffic-aware routing, admission control, traffic throttling, load shedding	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Understanding of working of various congestion control approaches
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
26	Quality of Service: Application Requirements, Traffic shaping, packet scheduling	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Basics of Quality of Service and understanding of traffic shaping and packet scheduling
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
27	Admission Control, Integrated Services, Differentiated Services	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Understanding of approaches such as Admission Control, Integrated Services and Differentiated Services to achieve QoS
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
28		Text Book	Computer Networks by Andrew S Tanenbaum

	Internetworking: Tunnelling, Internetwork Routing, Packet fragmentation	Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Understanding of challenges of internetworking and knowledge of approaches to overcome them
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
29	IPv4 header, Classes of IP addresses	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Basics of IPv4 addressing
30	Creation of subnets, classless addressing	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	How to create subnets and assign classless IPv4 addresses
31	Classless addressing, sub-netting in classless addressing, CIDR	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	How to create subnets for classless IP addressing and how to do Classless Inter Domain Routing
32	Concept of Network Address Translation, IPv6 header	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---

		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	How NAT works and understanding of IPv6 header
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
33	Other Network Layer Protocols: ICMP, IGMP, ARP, RARP	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Introduction to ICMP, IGMP, ARP and RARP protocols
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
34	Transport Service: Services provides, Service primitives, sockets, an example of socket programming	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Basics of Transport Layer
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
35	Elements of Transport Protocols: Addressing, Connection establishment and Release, Error Control	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Understanding of addressing, Connection establishment and release and Error control
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
36	Flow Control, Multiplexing, Crash Recovery, Congestion Control: Desirable bandwidth allocation	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board

		CLO	3,4
		SLO	Basics of flow control, multiplexing, crash recovery and congestion control at transport layer
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
37	Regulating the sending rate, wireless issues	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3,4
		SLO	How to regulate sending rate, Understanding challenges in wireless networks for transport layer
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
38	Connectionless Protocol: UDP header, Remote Procedure call, Realtime Transport Protocol	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Basics of connectionless protocols
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
39	Connection-oriented Protocol: Introduction to TCP, TCP header	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	Basics of TCP
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
40	TCP connection establishment and release, TCP Error Control	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3

		SLO	Understanding of TCP connection management and error control
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
41	TCP congestion control: slow start, congestion avoidance	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	2,3
		SLO	How TCP does congestion control
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
42	Performance Issues: Performance problems, performance measurement, host design for fast networks, fast segment processing, header compression	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	4
		SLO	Understanding of performance issues at transport layer and approaches to enhance performance
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
43	Domain Name System: Name space, Domain Resource Records, Name Servers	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3,4
		SLO	How DNS works
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
44	Electronic Mail: Architecture and Services, User Agent, Message Formats, Message Transfer, Final Delivery	Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3,4

45	Word Wide Web: Architecture, static web pages, dynamic web pages, HTTP, Mobile web, Web Search	SLO	How E-Mail Works
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446
		Text Book	Computer Networks by Andrew S Tanenbaum
		Reading (Reference)	---
		Pedagogy	PPT, Chalk and Board
		CLO	3,4
		SLO	How WWW works
		URL	https://lms.nirmauni.ac.in/course/view.php?id=9446

f. References

1. Andrew S. Tanenbaum, Computer Networks, PHI Publication
2. Behrouz Forouzan, Data Communication Networking, TMH Publication
3. Behrouz Forouzan, TCP/IP Protocol suite, TMH Publication
4. William Stallings, Data and Computer Communication, Pearson
5. Jim Kurose, Computer Networking: A top-down approach, Pearson

g. Instructions

Students are expected to carry out assigned work under Continuous Evaluation (CE) component and LPW component independently. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Academic integrity is expected from students in all components of course assessment.



CERTIFICATE

This is to certify that

Mr. / Ms. _____

Roll No. _____ *of Semester* _____

has satisfactorily completed the course in

Within the four walls of the
Institute.

Date of Submission: _____

Sign In-charge

Head of the Department

List of Experiments

Sr.	Title	Hours	CO
1	Understand following network commands with their utility: ping, ipconfig/ifconfig, iwconfig, netstat, traceroute/tracert, route, nslookup, hostname, dig, telnet, ssh, arp, nmap, getmac, pathping, ss and route add.	2	2,3
2	Experiments on packet capture tool: Wireshark To understand the features of Wireshark as a packet-capturing tool and encapsulation of information. Capture packets of following protocols HTTP/HTTPs, TCP, UDP, DNS, ICMP and ARP generated due to execution of earlier network commands and understand and analyze them.	2	1,2,3
3	Data link layer protocol: Implement simplex one-bit stop-and-wait protocol for noisy channels using socket programming in C language. Ensure to follow modular approach where physical layer implementation is done using simple socket connection between two machines and DLL protocol is implemented on top of it.	4	3
4	Wireless LAN: Configuration of Wireless LAN using cisco packet tracer using Access Point and Wireless Router with an objective to differentiate between these two devices. Also configure WLAN security mechanism using WEP, WPA, WPA2, MAC filtering etc.	4	3,4
5	Virtual LAN: Virtual LAN configuration using cisco packet tracer focusing on creating multiple VLANs and configuring inter-VLAN communication using Router on a stick method.	2	3,4
6	Internetworking with Routers: Design three or four simple networks using cisco packet tracer (with 3 to 4 hosts) and connect via Router. Understand the need to configure routing protocol and use static routing. Perform simulation and trace how routing is done in packet transmission. 1: Experiment on the same subnet. 2: Perform the experiment across the subnets and observe the functioning of the Router by selecting a suitable pair of Source and destination.	2	2,3
7	Subnetting: Design networks using Constant Length Subnet Masking (CLSM) and Variable Length Subnet Masking (VLSM) using cisco packet tracer with at least 4-5 subnets being created and interconnected. Assign static IP addresses across all subnets to understand the implementation of subnets. Repeat the same exercise by assigning the same IP addresses using DHCP server.	4	3,4
8	Dynamic Routing: Configure RIPv1, RIPv2 and OSPF on a network of at least 4-5 networks using cisco packet tracer. Design an appropriate topology to show the importance of these dynamic routing strategies.	4	3,4
9	File Transfer Client-Server: Develop a file transfer server supporting concurrent clients using socket programming in C. Also, develop the file transfer client to communicate with the server.	4	3
10	Configure Application Layer Protocols: Configure and demonstrate use of following application layer protocols: HTTP, DNS, FTP, SMTP, IMAP, POP3, DHCP etc using cisco packet tracer.	2	3
Total		30	

Additional Experiments

*A	Data Link Layer (Frame Generation): Write a program in C language to read a data stream from a data file to create frames by implementing the character stuffing concept and inserting control characters. Frames should be written in an intermediate file that acts as a channel. A separate receiving program should read frames from the intermediate file and decode received bytes, and write the decoded bytes to a separate file.	4	2,3
*B	NAT-PAT: Demonstrate Network Address Translation (NAT) and Port Address Translation (PAT) using CISCO Packet Tracer simulation.	2	3,4
*C	Experiment on Network Layer (Leaky Bucket/Token Bucket): Develop a program to implement a congestion control mechanism using either Leaky Bucket or Token Bucket technique in C language. The program must take the required input conditions to start the simulation, and the values must range up to a suitably high range to explain the exact behavior of the congestion control strategy.	4	2,3

Nirma University
Institute of Technology
Computer Science and Engineering Department

Practical Policy

B. Tech. (CSE), Int. B. Tech. CSE-MBA Semester - V Academic Year: 2026-27

Course Code & Name	:	3CS201CC24 Computer Networks
Credit Details	:	Lectures-3, Tutorial-0, Practicals-2, Credits-4
Course Co-ordinator	:	Dr. Pooja Chaturvedi
Contact No. & Email	:	9918596994, pooja.chaturvedi@nirmauni.ac.in
Office	:	Faculty Cabin W502A, 5 th floor, W-Block
Course Faculty:		
Dr Pooja Chaturvedi (Course Coordinator) Dr Vijay Ukani Dr Zunnun Narmawala Dr Umesh Bodkhe Dr Rajan Datta Dr Leena Patel Mohit Parmar		
Common Visiting Hours: Monday to Friday: 7:45 am to 3:00 pm Sat: 7:30 am to 1:30 pm Check Timetable for availability (https://nuweb.nirmauni.ac.in/tt/cse)		

Rubrics for Regular Evaluation Weightage (0.6)	Rubrics for Final Examination Weightage (0.4)
Active involvement in different tasks related to practical / Eagerness to learn	Demonstration of accurate understanding of the objective of practical
Regularity in attending practical sessions	Fundamental concepts and technical know-how about practical
Preparedness for practical session	Correlation of theoretical concepts with real-life applications
Ability to work in a team	Question-answers, writing and presentation skills
Originality, completeness, presentability and timely submission of assigned work/laboratory manual	Use of appropriate procedures, tools and techniques to conduct experiments and collect data
Discipline during practical session	

Criteria	Satisfactory (0.5 mark)	Adequate (1 mark)	Proficient (1.5 mark)	Outstanding (2 mark)
Lab participation (02 mark)	Student arrives late in the lab and rarely participate in lab proceedings. No attempt/desire to learn.	The student has a tendency to arrive late and unprepared. Such unpunctuality or unpreparedness makes it impossible to fully participate.	The student arrives on time to the lab, but may be unprepared and hence could not participate to the full extent.	The student arrives on time with complete preparation and participates in lab proceedings with full enthusiasm. Even eager to explain concepts to fellow batch mates and ready to assist them.
Definition of objectives and scope (02 mark)	The student is unaware of the practical objectives and concepts	The student has a difficulty in understanding/explaining key concepts of the practical	The student has a basic knowledge of content but may lack understanding of some of the concepts.	The student demonstrates an accurate understanding of the objectives and concepts.
Proper use of procedures (02 mark)	Unable to follow the instructions and perform the experiment.	Follows a limited set of instructions and performs the experiment half-heartedly.	Performs the experiment in a proper manner, however occasionally not following the procedures.	Follows all the instructions given by the instructor and performs the experiment in a perfect manner. Also, influence/emphasise others to follow the procedures.
Result analysis and discussion/ Timely Completion of the work (02 mark)	Calculations/Graphs/Quizzes are not complete and not submitted within the given deadline.	Calculations/Graphs/Quizzes are partially complete in a very random/haphazard or disorganised manner. Work is inaccurate and has a number of errors.	Calculations/Graphs/Quizzes are complete. However, student could do the work more neatly by incorporating all the required information.	Calculations/Graphs/Quizzes are complete and neat. They include all the required details like titles, sketches, units etc. Errors, if any are minimal
Question-Answer/Presentation (02 mark)	Unable to answer the questions. Poor language and communication with a number of mistakes.	Answers to the questions are basic and superficial suggesting that concepts are not fully grasped. Language and communication is not clear and fluent, suggesting scope for improvement	Questions are answered fairly well barring a few questions. The language is good. Communication is clear.	All the questions are answered completely and correctly. Language is error-free. Communication is clear and fluent. No grammatical mistakes.

Rubric for continuous assessment of practical (10 marks)

Note: Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Academic integrity is expected from students in all components of course assessment.

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Sr. No.	Title of experiment	Date	Marks (as per rubrics)					Total Marks (10)	Sign
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1.	Network Commands								
2.	Wireshark								
3.	Simplex one-bit stop and wait protocol for noisy channel								
4.	Wireless LANs configuration								
5.	Virtual LANs configuration								
6.	Internetworking with Routers								
7.	Static Routing with Subnetting								
8.	Dynamic Routing								
9	File Transfer Client-Server								
10.	Application Layer Protocols								

Lab Faculty Signature and Name